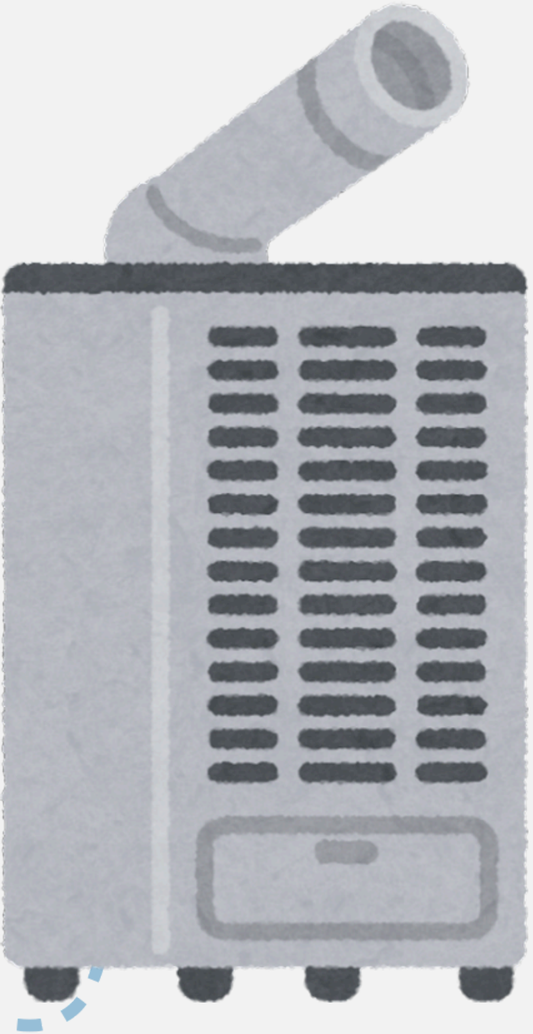
Team Name : Synergetix

# Shri Ram Product Challenge

## TerraCool - Reinventing the Air Cooler for India




**Team Member**

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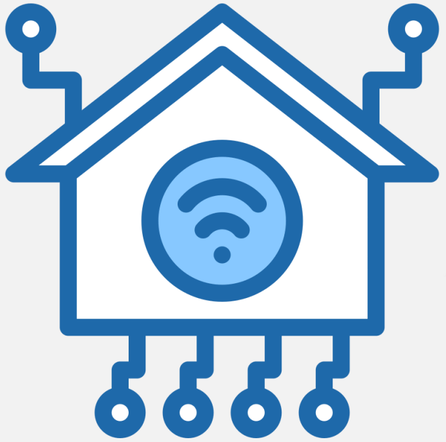
- Satyam Parida
- Samarth Jain

# The Opportunity

Large underserved mid-segment exists in India's cooling market; a ₹11–13K smart cooler could target ~1.5M households and reach break-even with only ~4% penetration

## India's Smart Cooling Gap

## India Air Cooler Market Size & Growth

**7.1%**

CAGR (2025-2033)

Source : IMARC Group

**$1.4Bn**

Revenue in the air coolers market

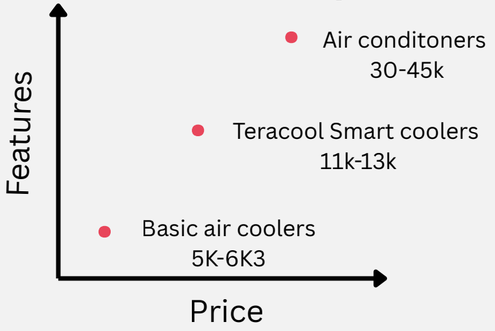
Source : IMARC Group

## COMPETITIVE ANALYSIS

| Company | Revenue | Key Smart Products | Price Range | Key Strength |
|---|---|---|---|---|
| **Symphony(Global Leaders)** | ₹1,200 Cr+ | BLDC Series, Touch Panel, i-Pure | ₹8k–₹35k | Global Scale, worlds #1 Cooler maker |
| **Bajaj Electricals** | ₹4,500 Cr+ | Limited smart range | ₹4k–₹15k | Mass brand trust; wide distribution |
| **Crompton Greaves** | ₹2,100 Cr+ | Select IoT models | ₹7k–₹20k | Strong in fans; expanding cooler range |
| **TerraCool (Current)** | ₹150–200 Cr | None – conventional only | ₹3k–₹10k | North/West India distribution depth |

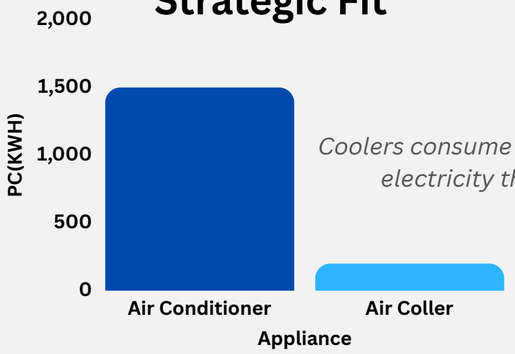
## Market Drivers

### Market Gap




**Gap identified**: Consumers seeking smart functionality without AC-level cost or electricity consumption currently have no option.

### Strategic Fit




*Coolers consume ~85–90% less electricity than ACs.*

### Opportunity Statement

Opportunity: Introduce a mid-priced smart air cooler (₹11–13K) combining affordability with smart features..

### Why TerraCool Can Capture This Opportunity?

- **Manufacturing expertise**

Existing air cooler production capability

- **Distribution network**

Strong Offline retail network + e-commerce channels

- **Brand positioning**

Trusted brand among middle-income households.

### Strength

Strong North & West India distribution network

Established brand equity in middle-income segment

### Weakness

Factory lacking IoT/Smart manufacturing capability

Lower brand recall compared to global leader Symphony

### Opportunity

PLI scheme incentives for domestic manufacturing

Massive gap between budget coolers and entry ACs

### Threat

Symphony launching competing BLDC smart coolers

Price-sensitive customers resisting 2x-3x premiums

### Recommendation Box

- Launch TerraCool Smart Air Cooler
- Target price: ₹11k–₹13k
- Target customers: Urban middle-income households

Prioritizing high-value, low-cost smart features allows TerraCool to deliver a modern cooling experience while maintaining affordability within the ₹11–13K target price range

# Feature Prioritization and Product Design

## Customer Value Drivers for Smart Cooling

### Convenience
App Based Control

### Comfort Optimization
Humidity Based Adjustment

### Energy Awareness
Real-Time Energy Tracking

### Affordability
Lower Power Cost than AC

## Feature Evaluation Criteria

Impact on user comfort and convenience

Hardware and software integration effort

Effect on final product price

## Feature Prioritization Matrix

**Customer Value** ↑ / **Implementation Cost** →

**Priority Features**
- Mobile App Control
- Humidity Sensor
- Energy Tracking
- Smart Timer

**Future Features**
- Voice Assistant Integration
- Air Purification

**Low Value/Low Cost**
- LED Display

**Low Value/High Cost**
- All Climate Optimization

## TerraCool Smart Cooler – V1

### Core Features
- Mobile App Control
- Humidity-based Auto Cooling
- Energy Usage Monitoring
- Smart Timer

### Positioning
Smart cooling without AC-level cost.

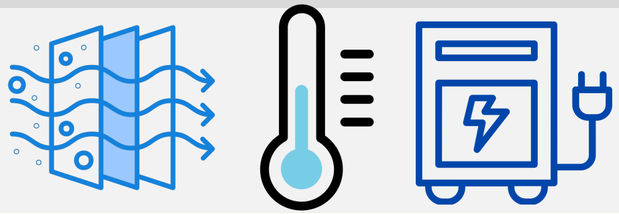
### Value Proposition
Modern,Affordable and Energy-Efficient for middle-income households

## Implementation Feasibility for TerraCool

### Hardware Feasibility
- ✓ Low-cost WiFi IoT module
- ✓ Single humidity sensor

### Manufacturing Feasibility
- ✓ Compatible with existing cooler architecture
- ✓ Minimal factory modifications

### Cost Feasibility
- ✓ Controlled manufacturing cost
- ✓ Enables ₹11–13K retail price



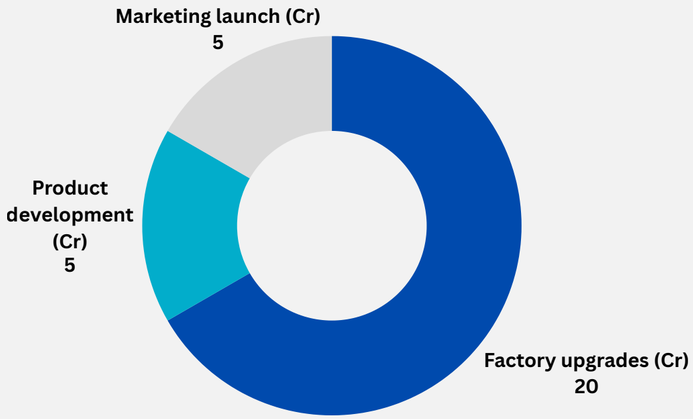
Source: Feature Prioritization based on customer needs , implementation feasibility , and cost considerations

A ₹30 Cr investment breaks even at ~60K units—requiring only ~4% penetration of the ~1.5M addressable market

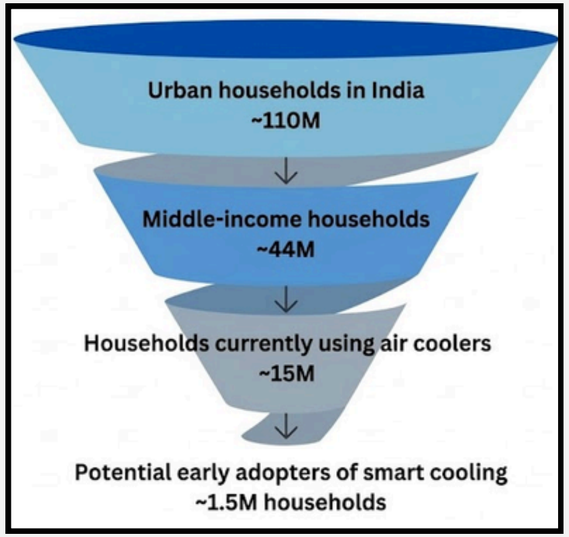
# Break-Even Analysis and Customer

## Key Assumptions

| Product Price | Production Cost | Initial Investment |
|---|---|---|
| **₹12K** Midpoint of ₹11–13K smart cooler range | **₹7K per unit** Includes cooling hardware + IoT module + Handeling cost | **₹30 Cr** Factory upgrade + development + marketing. |




**Factory upgrades account for ~67% of total investment, driven by IoT integration and production line upgrades**




**Assumption:** ~10% early adoption of smart cooling
Even ~4% penetration of the target market (~60K units) enables TerraCool to reach break-even.

## Unit Economics and Break-Even Calculation

**Investement Required 30Cr** → **Profit Per Unit 12K-7K = 5K** → **Break Even Units 60,000** → **Market Penetration needed 4%**

Financial Viability: Selling ~60K units enables TerraCool to recover it ₹30 Cr Iinvestment requiring only ~4% penetration of the addressable market

*Assumptions based on industry benchmarks for mid sized appliance manufacturers. Detailed calculations in Annexure*

## Personas

**RAHUL KUMAR** — The night-time waster

| Age | Salary | City | 1 |
|---|---|---|---|
| 32 | 28K | Jaipur | BHK |

Government clerk. Sleeps by 10:30pm but nobody remembers to switch the cooler off. Wakes up at 3am feeling cold, turns it off manually. Happens every night.

*"Raat ko cooler band karna yaad hi nahi rehta. Timer laga do toh subah chain ki neend aaye"*

**Feature he uses most**

**Smart timer — set once at bedtime, forget about it**

**Sunita Singh** — The Worried Mother

| Age | Salary | City | 2 |
|---|---|---|---|
| 38 | 35K | Kanpur | BHK |

Homemaker, two kids (8 & 11). Runs the cooler in the kids' room at night. Checks on them at midnight — often finds room too cold or kids coughing from dry air.

*"Bacchon ko subah uthke gala kharab hota hai cooler ki wajah se. Agar machine khud samajh le toh kitna acha ho"*

**Feature she uses most**

**Humidity auto-control + kids' room timer**

Targeting urban middle-income households through hybrid offline-online distribution and digital-led marketing enables TerraCool to rapidly scale smart cooler adoption in high-temperature markets

# Go-To-Market Strategy

## Target Customer Segment

### Customer Economics

✔ Household income ₹6L-₹15L annually

*Price sensitive segment that finds ACs expensive but can afford ₹11k-₹13k smart coolers*

### Climate Need

✔ Cities with high summer temperatures(>40°C)

- Delhi
- Jaipur
- Lucknow
- Ahemdabad

*These markets already have high cooler penetration*

### Technology Readiness

✔ Urban consumers with:

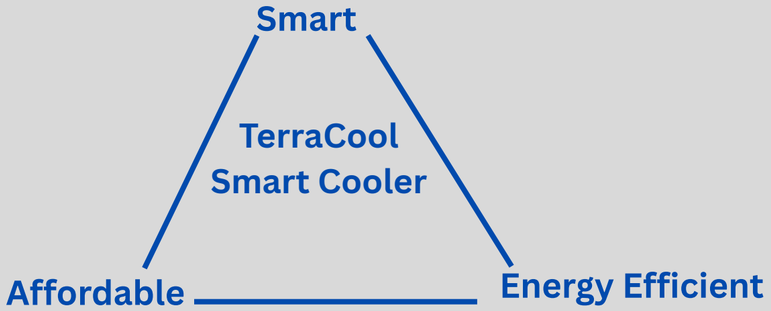
- Smartphone usage
- familarity with smart appliances

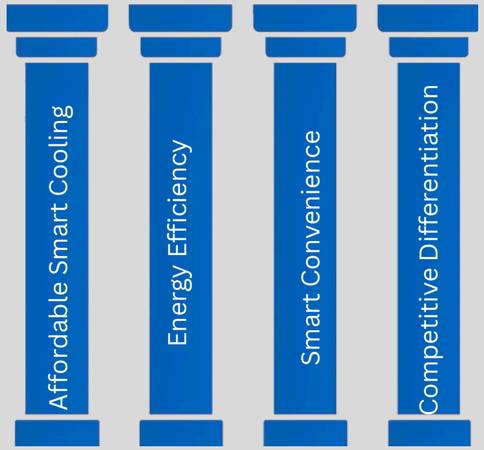
*This makes app controlled cooling attractive*

## Product Positioning

### Smart cooling without AC-level cost.

Affordable smart cooling designed for urban households seeking energy-efficient alternatives to ACs.



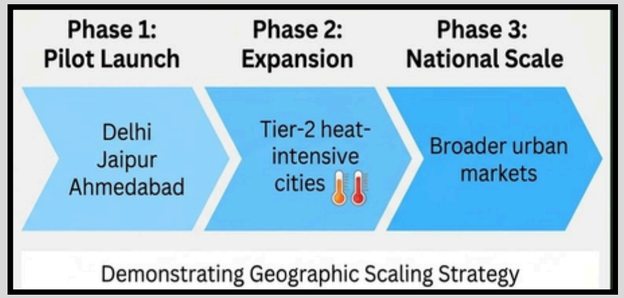

### VALUE PILLARS

- Price positioned at ₹11–13K, significantly lower than ₹30K+ entry-level ACs, making smart cooling accessible to middle-income households
- Consumes ~85–90% less electricity than ACs, helping reduce household energy bills during peak summer months.
- Mobile app control and humidity-based cooling adjustment enable automated and user-friendly cooling management.
- TerraCool bridges the gap between basic coolers and air conditioners by combining affordability, energy efficiency, and smart functionality.

## Channel Strategy

| Primary Channel | Supporting Channel |
|---|---|
| Offline Retail | E-commerce |
| Appliance Stores | Amazon/Flipkart |
| Regional Dealers | Brand Website |

*Offline channels drive trust and product demonstration, while e-commerce enables urban convenience purchases.*




| Retail Experience | Digital Marketing | Seasonal Campaigns |
|---|---|---|
| • In-store demonstrations<br>• Retailer incentives | • Targeted social media ads<br>• Search advertising before summer | • Pre-summer promotions<br>• Cooling season offers |

*Sources: Strategy based on Indian Cooling appliance usage patterns and urban adoption trends*

**Urban middle-income cooler households represent ~15M potential customers, creating an early smart cooler market opportunity of ~1.5M households**

# **Annexure A:** Market Structure & Demand Estimation

## MARKET STRUCTURE

| Product Category | Price Range | Key Characteristics |
|---|---|---|
| Basic Air Coolers | ₹5K–₹8K | Affordable but manual operation |
| Premium Air Coolers | ₹10K–₹15K | Improved cooling but limited smart functionality |
| Air Conditioners | ₹30K+ | Strong cooling but high upfront and electricity cost |

***The cooling market lacks a mid-priced product combining affordability with smart functionality, creating an opportunity for smart air coolers.***

## MARKET DEMAND ESTIMATION

| Step | Calculation | Result |
|---|---|---|
| Total households | Population ≈ 1.4B / Avg household size ≈ 4.5 | **≈ 310M** |
| Urban households | 310M × 35% urban share | **≈ 110M** |
| Middle-income households | 110M × 40% | **≈ 44M** |
| Cooler households | 44M × 35% cooler penetration | **≈ 15M** |
| Potential smart cooler adopters | 15M × 10% early adoption | **≈ 1.5M** |

### Key Assumptions

- Urban population share ≈ 35%
- Middle-income share ≈ 40%
- Cooler penetration ≈ 35%
- Early adoption of smart appliances ≈ 10%

## TARGET CUSTOMER SEGMENT

### Segment Profile

- Urban middle-income households
- Income Range: ₹6L-₹15L anuually
- Current cooling solutions are Fans and traditional air coolers

### Key Needs

- Affordable cooling solutions
- Lower electricity consumption
- Convenience through smart features

***This segment represents the largest group seeking better cooling without the cost of air conditioner***

## STRATEGIC MARKET INSIGHT

The **analysis** indicates a potential early market of **~1.5M** households for smart air coolers among urban cooler users.

Capturing even ~4% of this segment (~60K units) would enable **TerraCool** to achieve break-even, supporting the financial viability analysis.

**A ₹12k smart cooler with ~₹5k unit margin allows TerraCool to recover its ₹30 Cr investment after ~60k units**

# **Annexure B:** Product Design & Financial Assumptions

## FEATURE PRIORITIZATION FRAMEWORK

| Feature | Customer Value | Implementation cost |
|---|---|---|
| Mobile App control | High | Low |
| Humidity Sensor | High | Low |
| Smart Timer | Medium | Low |
| Energy Usage Tracking | Medium | Low |
| Voice Assistant | Medium | High |
| Air Purification | Medium | High |

***Features with high customer value and low implementation complexity were prioritized to maintain product affordability.***

## CORE PRODUCT CAPABILITIES

The selected features enable four key product capabilities.

### Smart Control

Mobile app-based operation allows users to control cooling remotely.

### Adaptive Cooling

Humidity sensor adjusts cooling performance based on environmental conditions.

### Automated Scheduling

Smart timer allows users to program cooling schedules.

### Energy Monitoring

Energy usage tracking helps users monitor electricity consumption.

***These capabilities deliver smart functionality without significantly increasing hardware complexity or manufacturing cost.***

## Manufacturing Cost Estimation

| Component | Estimated Cost | Justification |
|---|---|---|
| Cooling system & body | ₹3,500 | Standard cooler manufacturing benchmark |
| Cooling pads & pump | ₹1,200 | Typical evaporative cooler component cost |
| IoT module | ₹800 | Wi-Fi microcontroller and connectivity hardware |
| Humidity sensor | ₹200 | Standard environmental sensor pricing |
| Electronics & assembly | ₹1,300 | PCB, wiring, assembly and testing |

Total Manufacturing Cost: 7k

**Smart features increase production cost by ~₹1,000 compared to traditional coolers, enabling differentiation while maintaining competitive pricing.**

## Investment

### Investment Breakdown

| Category | Investment | Purpose |
|---|---|---|
| Production line upgrades | ₹8 Cr | Smart component assembly capability |
| Electronics integration setup | ₹5 Cr | PCB testing and calibration systems |
| Tooling & molds | ₹4 Cr | New product casing and tooling |
| Quality testing infrastructure | ₹3 Cr | Reliability testing and quality control |

***Given a ~1.5M household potential market, TerraCool needs to capture ~4% market share to recover its investment.***

# Thank you!